

ILLUSTRATIVE COURSE • CASE PACK

Community learning-support indicators

A small fictional dataset for data-quality, interpretation and decision-making practice.

A worked model—not a prescribed course

Every field is filled to show the expected level of clarity. Facilitators should adapt the evidence, content, dates and assessment to their own discipline and obtain academic approval before delivery.

Teaching example • Not operational UCC data

Safe practice data

All records and values in this pack are fictional. They must not be interpreted as University of Cape Coast performance data or used for operational decisions.

Scenario

A microcredential team wants to decide which learner-support channel should receive additional attention next month. Six weekly summaries have been supplied. You must inspect data quality, compare demand and service performance, and recommend an action without claiming more than the dataset supports.

Fictional weekly data

Week	Channel	Requests	Resolved	Median min	Satisfaction /5
1	Email	84	70	310	3.7
2	Live chat	102	93	18	4.3
3	Telephone	61	52	42	4.0
4	Email	97	76	355	3.5
5	Live Chat	116	108	17	4.4
6	Telephone	58	—	46	3.8

Notice the spelling variation in Live chat / Live Chat and the missing resolved value in Week 6. Do not silently repair either issue; record the rule and consequence in your cleaning log.

Data dictionary

Variable	Definition	Type / valid values
week	Sequential reporting week in this teaching extract.	Integer 1–6
channel	Primary support route used by learners.	Email, Live chat, Telephone
requests	New support requests logged during the week.	Non-negative integer
resolved	Requests marked resolved by the reporting cut-off.	Non-negative integer; may be missing
median_minutes	Median elapsed minutes to first substantive response.	Positive number; minutes
satisfaction	Mean response to a five-point post-service item.	Number 1.0–5.0

Required analysis

1. State the decision question, unit of analysis and limits of the six-row extract.
2. Check missing values, category consistency, valid ranges and whether resolved can exceed requests.
3. Create a transparent cleaned version while retaining the original data.
4. Calculate resolution rate only where the inputs are available; state how missing data is handled.
5. Create one suitable chart comparing a meaningful indicator by channel.
6. Write one finding, at least two limitations and a proportionate recommendation.

Questions for critical interpretation

- Does high request volume necessarily indicate poor service? What other explanations are plausible?
- Can satisfaction scores be compared without knowing the response rate and question wording?
- Could requests opened in one week be resolved in a later week? How would that affect a resolution rate?
- What disaggregation or contextual evidence is needed before resources are reallocated?
- What personal or sensitive data should be excluded from a public notebook?

Facilitator annotation

A strong learner response cleans only what can be justified, shows all assumptions, distinguishes observation from interpretation and recommends further verification where the evidence is weak.